

Implementation of Machine learning models for "El Pino" Hospital

BENJAMIN SALAZAR¹

¹ Andres Bello University, Department of Engineering, Chile (e-mail: b.salazarallendes@uandresbello.edu)

Corresponding author: Benjamin Salazar (e-mail: b.salazarallendes@uandresbello.edu).

I. INTRODUCTION

THIS document will address the use and importance of Diagnosis-Related Groups (DRG) in hospitals and how good management of the patients information can improve management of resources and improve clinical decision making. DRG are a system in which patients are classified according to their clinical information, such as their diagnoses, procedures, age, and sex.

Due to the large number of data that has to be managed, manual classification can be complex, slow, and prone to mistakes. Because of this, the use of Machine Learning allows hospitals to automate this process to speed classification and improving consistency.

The work in this document proposes different Machine Learning models to study the efficiency of each one in classifying DRG for hospitals.

II. RELATED WORK

The classification of patients is a key element in health systems because it allows to optimize resources and organize medical attention based on the severity of the patients condition. International Organizations highlight the great value of efficient patient's clinical information management.(OECD,2022)

In recent years implementation of machine learning techniques has been increasing significantly in health institutions, Allowing hospitals to analyze large amounts of data and developing predictive models that help in clinical decision making (Shickel et al., 2021)

Specifically, medical text data has been broadly studied since much of the clinical information is unstructured, and many machine learning models have been proven to be effective (Zhang et al., 2022)

Studies report that the use of Logistic regression and Naive Bayes in text classification show good results (Kowsari et al.,2021)

This information proves that the implementation of machine learning algorithms in clinical data allows better classification processes.

III. OBJECTIVE

The objective of this study is to compare the efficiency and accuracy of two different machine learning algorithms in classifying patients in DRG groups for the hospital "El Pino", Logistic Regression and Naive Bayes.

IV. METHODOLOGY

A. DATASET DESCRIPTION

The dataset from Hospital El Pino contains patients clinical data, including:

- Diagnoses
- Procedures
- Age
- Sex
- DRG

Since some patients have multiple diagnoses or procedures, diagnoses and procedures have been each one rewritten in a single text row, to show only relevant information to the machine learning model.

B. DEVELOPMENT METHODOLOGY

1) Model development process

The process of the development in this work is as follows:

- Preparing and tidying data
- Convert text using TF-IDF
- Separation of the dataset in a training set equal to 80% of the data and a test set equal to 20% of the data
- Training of the model
- evaluating performance of the model
- comparing performance of both models

2) Machine learning models

two machine learning models were chosen for this task:

- Logistic Regression
- Naive Bayes

These models were chosen for their adequacy to work with text data, and each model was modified several times, changing parameters in each version, to test how the results will change with different iterations of the models.

3) Assessment metrics

the following metrics were chosen to evaluate the performance of the models:

- Accuracy
- Precision
- Recall
- F1-score

V. EXPLORATORY DATA ANALYSIS

A. DATA QUALITY

An analysis of data quality was made taking into consideration:

- Completeness: No null values were found in the variable.
- Correctness: A wide range of ages was found.
- Outliers: One outlier value was found in the age variable.

B. DESCRIPTIVE STATISTICS

Descriptive statistics were calculated with the age variable. Results show a wide distribution in patients age, from newborns to senior patients, the median being patients the adulthood around 35 to 40 years. A large number of newborn patients are being treated. On the other side, there is a small number of senior patients over the age of 80.

	age
count	14561
mean	39.42655037
std	24.68154549
min	0
25%	23
50%	36
75%	60
max	121

TABLE 1. Age analysis results

C. DATA VISUALIZATION

Graphs were made in order to visualize relevant information:

- Age box plot
- Age distribution chart
- DRG distribution chart

The analysis shows an important disparity in the DRG classes, having some DRG's with a great number of patients, while others have very few.

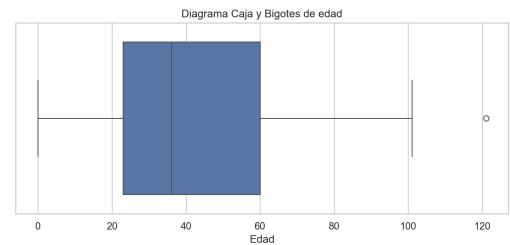


FIGURE 1. Box plot showing age

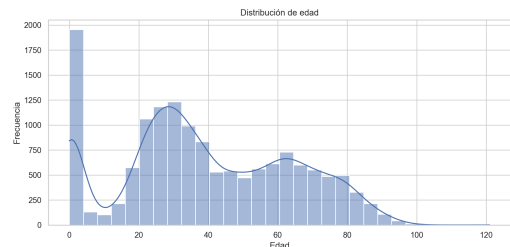


FIGURE 2. Plot showing patients age

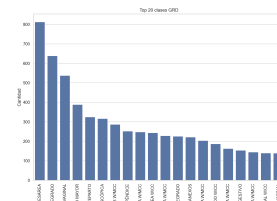


FIGURE 3. Plot showing the patient distribution of the DRG groups

VI. EXPERIMENTS AND RESULTS

A. INPUT VALUE SELECTION

Input values for the machine learning model correspond to Diagnostics and Procedures, meanwhile the output will be the DRG for the patient.

B. MODEL RESULTS

1) Logistic Regression results

This model was tested with different configurations, showing similar results, meaning there is not much room to improvement without changing the dataset. Results are shown in the following table:

Criteria	v1	v2	v3	v4	v5	v6
Accuracy	0.55	0.50	0.50	0.51	0.55	0.46
F1-score	0.46	0.51	0.51	0.51	0.55	0.49
weighted Precision	0.46	0.60	0.60	0.60	0.63	0.64
macro Precision	0.16	0.26	0.26	0.26	0.35	0.37
weighted Recall	0.55	0.51	0.51	0.52	0.56	0.46
macro Recall	0.17	0.30	0.30	0.29	0.41	0.31

TABLE 2. Logistic Regression result table

The data in the table show how in every version of the model results are similar, averaging 0.52 Accuracy and F1-

score. The fifth version is the one with better performance, although the improvement is marginal. Its also important to note that the sixth version took significantly longer to show the results, reaching the max of iterations.

2) Naive Bayes results

This model was treated the same way as the first model, starting with a base version and changing parameters to analyze performance through the versions. Results are shown in the following table:

Criteria	v1	v2	v3	v4	v5
Accuracy	0.35	0.50	0.39	0.31	0.58
F1-score	0.24	0.41	0.28	0.21	0.55
weighted Precision	0.22	0.42	0.30	0.19	0.57
macro precision	0.05	0.18	0.11	0.04	0.29
weighted recall	0.35	0.51	0.40	0.32	0.58
macro recall	0.06	0.16	0.09	0.05	0.29

Naive Bayes showed an important downgrade in performance, obtaining 0.35 Accuracy and 0.24 F1-score, but when changing the alpha value of the model to 0.1 in version 2, the results were almost similar to Logistic regression. posterior versions with greater alpha values lowered the performance, and a final test was made with an alpha value of 0.1 and changing the text vectorization technique from TF-IDF to CountVectorizer. This change achieved better performance than Logistic regression.

C. MODELS ARCHITECTURE

The architecture of the two models studied in this experiment were:

- Logistic Regression with TF-IDF
- Naive Bayes evaluated with TF-IDF and Count Vectorization

D. PERFORMANCE ANALYSIS

1) Logistic Regression performance

Confusion matrix analysis shows that the Logistic regression model performance is good in frequent DRG classes, but has problems classifying scarce DRG classes. This information tells us that this model is adequate for classes with a solid and even amount of data, is the number of data lowers, the performance drops.

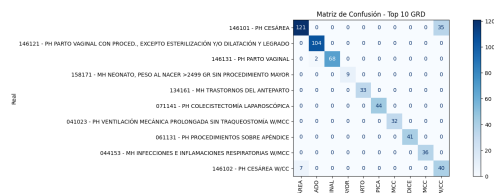


FIGURE 4. Confusion matrix of the top 10 most common DRG classes for Logistic Regression

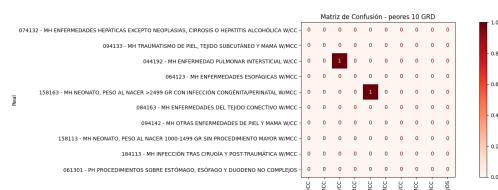


FIGURE 5. Confusion matrix of the top 10 least common DRG classes for Logistic Regression

2) Naive Bayes performance

Confusion matrix analysis shows that Naive Bayes has a similar performance to Logistic Regression in the most common DRG classes, with some misclassifications, but has a worse performance with the scarce DRG classes. This, along with the low Accuracy and F1-score shows a significantly lower performance.

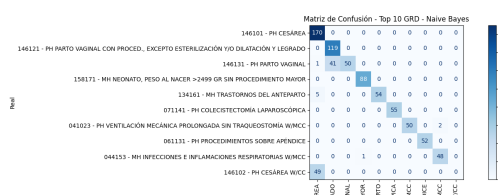


FIGURE 6. Confusion matrix of the top 10 most common DRG classes for Naive Bayes

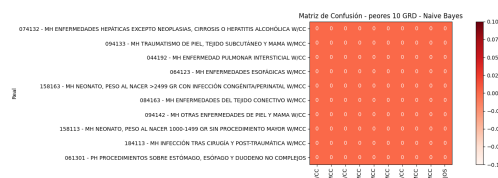


FIGURE 7. Confusion matrix of the top 10 least common DRG classes for Naive Bayes

but when changing the alpha value of the model to 0.1, and implementing Count Vectorization, Naive Bayes achieves a better performance than Logistic Regression. Overall, the performance is better, but when analyzing the least common DRG classes, the performance is worse than Logistic Regression.

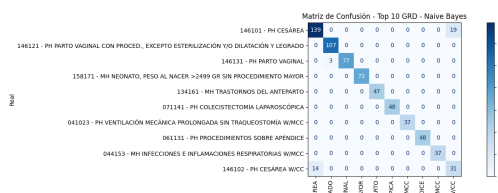


FIGURE 8. Confusion matrix of the top 10 most common DRG classes for Naive Bayes, using Alpha 0.1 and count vectorization

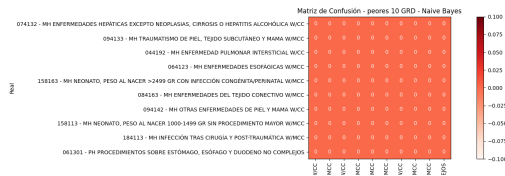


FIGURE 9. Confusion matrix of the top 10 least common DRG classes, using Alpha 0.1 and count vectorization

E. BIBLIOGRAPHY COMPARISON

Results obtained in this work are in line with studies done previously, where a performance offset can be observed in the lower patient populated DRG classes.

VII. CONCLUSION

Logistic regression model shows an average performance, correctly classifying DRG classes with a high number of patients but having problems when classifying DRG classes with low number of patients. Naive Bayes model shows a significantly lower performance, correctly classifying DRG classes with a high number of patients, but when the number lowers it starts classifying incorrectly, and with the least populated classes, there were almost no classifications. Changing the parameters of Naive Bayes made it have better performance than Logistic regression, but the performance in the scarce DRG classes did not change. The performance shown by these models lies in the imbalance in the data set rather than in the models themselves.

VIII. LIMITATIONS AND FUTURE WORK

This work had several limitations:

- Strong class imbalance
- Limited representation of rare classes
- Use of simple models

Future work will cover:

- Implementation and analysis of new machine learning models to improve the classification of scarce classes

IX. GITHUB REPOSITORY LINK

All the files used for this work are in the following Github repository:

- <https://github.com/BenjaminSalazarA/Clasificaci-n-DRG-elpino>

X. REFERENCES

- OECD. (2022). Health at a Glance 2022: OECD Indicators.
- Shickel, B., Tighe, P., Bihorac, A., & Rashidi, P. (2021). Deep learning in electronic health records.
- Zhang, Y., Chen, Y., & Wang, L. (2022). Medical text classification using machine learning techniques.
- Kowsari, K., et al. (2021). Text classification algorithms: A survey.